

Generation Interconnection System Impact Study Report

For Queue Project Q18-1107

Gallows Road 138 kV

May 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1107. The Interconnection Customer proposes a 129.1 MW solar photovoltaic facility interconnecting at the Gallows Road 138 kV substation in the Virginia area. The request was received on November 21, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 370 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,257,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Holmes Crk - Gallows Rd 138 KV	Idylwood - Gallows Rd 138 KV line	118.1	219	55.5
New Sterling - Cia Substation 138 KV	Cia - Cowlitz Falls 138 KV	108.4	327	12.7
Cia Substation - Papago Buttes 138 KV Ckt 1	Papago Buttes - Cia 138 KV Line 2	106.2	368	12.2
Cia - Papago Buttes 138 KV L	Cia - Papago Buttes 138 KV Cir. 1	106.2	368	12.2
Mottman - Papago Buttes 138 KV Line 1	Mottman Substation - Papago Buttes 138 KV Line	105.1	410	12.8
Mottman - Papago Buttes 138 KV Ckt 2	Mottman Substation - Papago Buttes 138 KV #1	105.1	410	12.8
Cowlitz Falls - Cia 138 KV	Cia - New Sterling 138 KV	104.3	257	7.8
Falls Church - Idylwod 138 KV Line	Annandale - Centennial 138 KV	102.9	256	43.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Holmes Crk - Gallows Rd 138 KV	Rebuild line with higher-capacity conductor	\$1,349,000	39% (\$526,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New Sterling Substation - Cia 138 KV L	Reconductor line	\$807,000	21% (\$172,000)
Papago Buttes - Cia 138 KV line	Reconductor line	\$470,000	63% (\$295,000)
Cia - Papago Buttes 138 KV Cir. 2	Reconductor line	\$552,000	63% (\$346,000)
Mottman - Papago Buttes 138 KV Line 1	Reconductor line	\$202,000	21% (\$43,000)
Mottman - Papago Buttes 138 KV Ckt 2	Reconductor line	\$468,000	21% (\$100,000)
Cia - Cowlitz Falls 138 KV	Reconductor line	\$830,000	63% (\$521,000)
Falls Church - Idylwood 138 KV Line	Reconductor line	\$588,000	43% (\$254,000)
	Total allocated to project		\$2,257,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.